

Hyungjin Chung, Ph.D.

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Research interests Generative models, Alignment, Imaging, AI for medicine

Work Experience

Korea University	Seoul, Korea
Assistant Professor, Dept. of CSE	2026.09 — Current
EverEx	Seoul, Korea
Director of Resesarch	2026.09 — Current
Lead AI Research Scientist	2024.08 — 2026.08
NVIDIA Research	San Jose, USA (remote)
Research Scientist Intern, AI4Science	2023.11 — 2024.01
Google Research	Mountain View, USA
Student Researcher, team LUMA	2023.07 — 2023.10
Los Alamos National Laboratory	Los Alamos, USA
Research intern, Applied math & Plasma physics	2022.06 — 2022.08

Education

KAIST	Daejeon, Korea
Ph.D., Bio & Brain Engineering	2019.03 — 2025.02
Advisor: Jong Chul Ye	
Thesis: <i>Practical approximations of posterior sampling in diffusion model-based inverse problems</i>	
Korea University	Seoul, Korea
B.S., Biomedical Engineering	2015.03 — 2019.02

Conf. publications *First authors, [†]Corresponding authors. If absent, first/last authors are first/corresponding.

[C29] Accelerating Video Inverse Problem Solvers with Autoregressive Diffusion Models
Taesung Kwon*, Jonghyun Park*, [Hyungjin Chung](#)[†], Jong Chul Ye[†]
NeurIPS 2026

[C27] Stitched Value Model for Diffusion Alignment
Hyojun Go, [Hyungjin Chung](#)[†], Prune Truong, Goutam Bhat, Li Mi, Zhaochong An, Zixiang Zhao[†], Dominik Narnhofer, Serge Belongie, Federico Tombari, Konrad Schindler
NeurIPS 2026

[C26] Exact Posterior Score Estimation for Solving Linear Inverse Problems
Abbas Mammadov, Ozgur Kara, Kaan Oktay, Iskander Azangulov, Adil Kaan Akan, [Hyungjin Chung](#), James Matthew Rehg, Yee Whye Teh
NeurIPS 2026

[C25] Generating Human Motion Videos using a Cascaded Text-to-Video Framework
Hyelin Nam, Hyojun Go, Byeongjun Park, Byung-Hoon Kim[†], [Hyungjin Chung](#)[†]
BMVC 2026

Yeobin Hong*, Suhyeon Lee*, [Hyungjin Chung](#)[†], Jong Chul Ye[†]
ECCV 2026

[C24] ReDesign: Recovering Editable Design Structures from Raster Images via Agentic Decomposition
Jooyeol Yun, Jintae Park, Hyesu Lim, Junha Hyung, [Hyungjin Chung](#), Jaegul Choo
ECCV 2026

- [C23] ContrastiveCFG: Guiding Diffusion Sampling by Contrasting Positive and Negative Concepts
Jinho Chang, Changsun Lee, [Hyungjin Chung](#), Jong Chul Ye
ICML 2026
- [C22] ReDirector: Robust Video Diffusion with Reversible Frame Reordering
Byeongjun Park, Byung-Hoon Kim, [Hyungjin Chung](#)[†], Jong Chul Ye[†]
CVPR 2026
- [C21] Video Parallel Scaling: Aggregating Diverse Frame Subsets for VideoLLMs
[Hyungjin Chung](#), Hyelin Nam, Jiyeon Kim, Hyojun Go, Byeongjun Park, Junho Kim, Joonseok Lee, Seongsu Ha, Byung-Hoon Kim
CVPR 2026 Findings
- [C20] A Foundational Brain Dynamics Model via Stochastic Optimal Control
Joonhyeong Park*, Byoungwoo Park*, Chang-Bae Bang, Jungwon Choi, [Hyungjin Chung](#), Byung-Hoon Kim[†], Juho Lee[†]
ICLR 2026
- [C19] InvFusion: Bridging Supervised and Zero-shot Diffusion for Inverse Problems
Noam Elata*, [Hyungjin Chung](#)*, Jong Chul Ye, Tomer Michaeli, Michael Elad
NeurIPS 2025
- [C18] Lesion-Aware Post-Training of Latent Diffusion Models for Synthesizing Diffusion MRI from CT Perfusion
Junhyeok Lee, Hyungwoong Kim, [Hyungjin Chung](#), Heeseong Eom, Jang Joon, Chul-Ho Sohn, Kyu Sung Choi
MICCAI 2025
- [C17] CapeLLM: Support-Free Category-Agnostic Pose Estimation with Multimodal Large Language Models
Junho Kim, [Hyungjin Chung](#)[†], Byung-Hoon Kim[†]
ICCV 2025
- [C16] VideoRFSplat: Direct Scene-Level Text-to-3D Gaussian Splatting Generation with Flexible Pose and Multi-View Joint Modeling
Hyojun Go*, Byeongjun Park*, Hyelin Nam, Byung-Hoon Kim, [Hyungjin Chung](#)[†], Changick Kim[†]
ICCV 2025
- [C15] SteerX: Creating Any Camera-Free 3D and 4D Scenes with Geometric Steering
Byeongjun Park*, Hyojun Go*, Hyelin Nam, Byung-Hoon Kim, [Hyungjin Chung](#)[†], Changick Kim[†]
ICCV 2025
- [C14] Derivative-Free Diffusion Manifold-Constrained Gradient for Unified XAI
Won Jun Kim*, [Hyungjin Chung](#)*, Jemin Kim*, Byeongsu Sim, Sangmin Lee, Jong Chul Ye
CVPR 2025
- [C13] CFG++: Manifold-constrained Classifier Free Guidance for Diffusion Models
[Hyungjin Chung](#)*, Jeongsol Kim*, Geon-Yeong Park*, Hyelin Nam*, Jong Chul Ye
ICLR 2025
- [C12] Regularization by texts for latent diffusion inverse solvers
Jeongsol Kim*, Geon-Yeong Park*, [Hyungjin Chung](#), Jong Chul Ye
ICLR 2025 (spotlight)
- [C11] Deep Diffusion Image Prior for Efficient OOD Adaptation in 3D Inverse Problems
[Hyungjin Chung](#) and Jong Chul Ye
ECCV 2024
- [C10] Prompt-tuning Latent Diffusion Models for Inverse Problems

[Hyungjin Chung](#), Jong Chul Ye, Peyman Milanfar, Mauricio Delbracio

ICML 2024

[C9] Decomposed Diffusion Sampler for Accelerating Large-Scale Inverse Problems

[Hyungjin Chung](#), Suhyeon Lee, Jong Chul Ye

ICLR 2024

[C8] Direct Diffusion Bridge using Data Consistency for Inverse Problems

[Hyungjin Chung](#), Jeongsol Kim, Jong Chul Ye

NeurIPS 2023

[C7] Improving 3D Imaging with Pre-Trained Perpendicular 2D Diffusion Models

Suhyeon Lee*, [Hyungjin Chung](#)*, Minyoung Park, Jonghyuk Park, Wi-Sun Ryu, Jong Chul Ye

ICCV 2023

[C6] Score-based Diffusion Models for Bayesian Image Reconstruction

Michael T. Mccann, [Hyungjin Chung](#), Jong Chul Ye, Marc L. Klasky

ICIP 2023

[C5] Parallel Diffusion Models of Operator and Image for Blind Inverse Problems

[Hyungjin Chung](#)*, Jeongsol Kim*, Sehui Kim, Jong Chul Ye

CVPR 2023

[C4] Solving 3d inverse problems using pre-trained 2d diffusion models

[Hyungjin Chung](#)*, Dohoon Ryu*, Michael T. Mccann, Marc L. Klasky, Jong Chul Ye

CVPR 2023

[C3] Diffusion Posterior Sampling for General Noisy Inverse Problems

[Hyungjin Chung](#)*, Jeongsol Kim*, Michael T. Mccann, Marc L. Klasky, Jong Chul Ye

ICLR 2023 (Notable-top-25%)

[C2] Improving Diffusion Models for Inverse Problems using Manifold Constraints

[Hyungjin Chung](#)*, Byeongsu Sim*, Dohoon Ryu, Jong Chul Ye

NeurIPS 2022

[C1] Come-Closer-Diffuse-Faster: Accelerating Conditional Diffusion Models for Inverse Problems through Stochastic Contraction

[Hyungjin Chung](#), Byeongsu Sim, and Jong Chul Ye

CVPR 2022

Journal publications

[J15] Deep Learning for Deep Learning Performance: How Much Data Is Needed in Biomedical Imaging?

Kyu Sung Choi, Junhyeok Lee, [Hyungjin Chung](#), Jeong-Hoon Lee

PLOS One, 2026

[J14] Label-independent Framework for Objective Evaluation of Cosmetic Outcome in Breast Cancer

Sangjoon Park, Yong Bae Kim, Jee Suk Chang, Seo Hee Choi, [Hyungjin Chung](#), Ik Jae Lee, Hwa Kyung Byun

Artificial Intelligence in Medicine, 2025

[J13] Steerable Conditional Diffusion for Out-of-Distribution Adaptation in Medical Image Reconstruction

Alexander Denker*, Riccardo Barbano*, [Hyungjin Chung](#)*, Tae Hoon Roh, Simon Arridge, Peter Maass, Bangti Jin, Jong Chul Ye

IEEE TMI, 2025

[P2] Objective and Interpretable Breast Cosmesis Evaluation with Attention Guided Denoising Diffusion Anomaly Detection Model

Sangjoon Park, Yong Bae Kim, Jee Suk Chang, Seo Hee Choi, [Hyungjin Chung](#), Ik Jae Lee, Hwa Kyung Byun

IjROBP, 2024

- [J12] Fundus image enhancement through direct diffusion bridges
Sehui Kim*, [Hyungjin Chung*](#), Se Hie Park, Eui-Sang Chung, Kayoung Yi, Jong Chul Ye
IEEE JBHI, 2024
- [J11] MR Image Denoising and Super-Resolution Using Regularized Reverse Diffusion
[Hyungjin Chung](#), Eun Sun Lee, Jong Chul Ye
IEEE TMI, 2022
- [J10] Low-dose sparse-view HAADF-STEM-EDX tomography of nanocrystals using unsupervised deep learning
Eunju Cha*, [Hyungjin Chung*](#), Jaeduck Jang, Junho Lee, Eunha Lee, Jong Chul Ye
ACS Nano, 2022
- [J9] Score-based diffusion models for accelerated MRI
[Hyungjin Chung](#) and Jong Chul Ye
Medical Image Analysis, 2021
- [J8] Unsupervised Deep Learning Methods for Biological Image Reconstruction and Enhancement
Mehmet Akçakaya, Burhaneddin Yaman, [Hyungjin Chung](#), Jong Chul Ye
IEEE SPM, 2021
- [J7] A Deep Learning Model for Diagnosing Gastric Mucosal Lesions Using Endoscopic Images: Development, Validation, and Method Comparison
Joon Yeul Nam*, [Hyungjin Chung*](#), Kyu Sung Choi*, Hyuk Lee* et al.
Gastrointestinal Endoscopy, 2021
- [J6] Feature Disentanglement in generating three-dimensional structure from two-dimensional slice with sliceGAN
[Hyungjin Chung](#), Jong Chul Ye
Nature Machine Intelligence, 2021
- [J5] Missing Cone Artifacts Removal in ODT using Unsupervised Deep Learning in Projection Domain
[Hyungjin Chung*](#), Jaeyoung Huh*, Geon Kim, Yong Keun Park, Jong Chul Ye
IEEE TCI, 2021
- [J4] Two-Stage Deep Learning for Accelerated 3D Time-of-Flight MRA without Matched Training Data
[Hyungjin Chung](#), Eunju Cha, Leonard Sunwoo, Jong Chul Ye
Medical Image Analysis, 2021
- [J3] Deep learning STEM-EDX tomography of nanocrystals
Yoseob Han*, Jaeduck Jang*, Eunju Cha*, Junho Lee*, [Hyungjin Chung*](#) et al.
Nature Machine Intelligence, 2021 (March Issue cover)
- [J2] Unpaired training of deep learning tMRA for flexible spatio-temporal resolution
Eunju Cha, [Hyungjin Chung](#), Eung Yeop Kim, Jong Chul Ye
IEEE TMI, 2021
- [J1] Unpaired deep learning for accelerated MRI using optimal transport driven cycleGAN
Gyutaek Oh, Byeongsu Sim, [Hyungjin Chung](#), Leonard Sunwoo, Jong Chul Ye
IEEE TCI, 2020

Books

- [B2] Generative Machine Learning Models in Medical Image Computing
Chapter 7: Diffusion Models for Inverse Problems in Medical Imaging
[Hyungjin Chung](#), Jong Chul Ye
- [B1] Deep Learning for Biomedical Image Reconstruction
Chapter 12: Image Synthesis in Multi-Contrast MRI with Generative Adversarial Networks
Tolga Çukur, Mahmut Yurt, Salman Ul Hassan Dar, [Hyungjin Chung](#), Jong Chul Ye

Awards	31st Samsung Humantech Silver Award	2025.02
	Google Conference Scholarship	2024.05
	30th Samsung Humantech Gold Award	2024.02
	29th Samsung Humantech Gold Award	2023.02
	2020-2024 BISPL Best Researcher Award	2020-2024.12
Professional Service	Area Chair	
	ICLR	2027-
	Committee Member	
	KSIAM Imaging Science	2026-2027
	Technical Group on Image Processing, IEIE	2026-
	Grant Reviewer	
	The Royal Society (Leverhulme Trust Senior Fellowship)	2026
	Reviewer (Conference)	
	ICML (<i>Gold reviewer 2026</i>)	2024-
	ICLR	2024-
	NeurIPS	2022-
	NeurIPS Datasets&Benchmarks	2023-
	CVPR	2023-
	ECCV	2022-
	ICCV	2023-
	AAAI	2025-
	SIGGRAPH	2026-
	SIGGRAPH Asia	2025-
	MICCAI	2022-
	Reviewer (Journal)	
	NEJM AI	
	Nature Communications	
	npj Digital Medicine	
	Medical Image Analysis	
	TMLR	
	IEEE TMI (<i>Gold Distinguished reviewer 2024, Bronze Distinguished reviewer 2023</i>)	
	IEEE TPAMI, TCI, TSP, TIP, SPS, SPL, TRPMS	
	See full list	
Invited talks	Towards Controllable Video World Models	
	- <i>Conference on Artificial Intelligence and Signal Processing</i>	2026.09
	Closing the loop of alignment in Generative AI	
	- <i>KAIST EE</i>	2026.05
	- <i>Spring Workshop on Broadcasting and Media Technology, KIBME</i>	2026.05
	Towards exact posterior sampling with diffusion models	
	- <i>KSIAM</i>	2026.05
	Physically Plausible Generative AI	
	- <i>KAIST AI</i>	2026.03
	- <i>DGIST AI</i>	2026.03
	- <i>ETH-PRS</i>	2026.02
	- <i>Joint Winter School on Image Understanding and Image Processing by IEIE</i>	2026.01
	Towards motion understanding and generation in videos	
	- <i>Dept. of Applied Statistics and Data Science, Yonsei University</i>	2025.10

- Korean Society of Digital Health	2025.09
Texts in inverse problem solving using diffusion models	
- University of Michigan	2024.10
Tutorial on Denoising Diffusion Model: Fundamentals & Applications	
- IEIE: Winter School on Biomedical Signal Processing	2024.02
Adapting diffusion models for inverse problems	
- UCLA, Caltech: Grundfest Memorial Lecture Series in Graphics and Imaging	2024.02
- 2023 NeurIPS Workshop on diffusion models	2023.12
- Google Research	2023.10
Advances in diffusion models and their applications to inverse problems	
- Guest Lecture, Korea University	2023.11
Generative (diffusion) models for medical imaging	
- KoSAIM 2025 summer school	2025.08
- International Congress on Magnetic Resonance Imaging (ICMRI) 2023	2023.11
- Michigan State University	2023.09
- Stanford MedAI	2023.08
- MGH, School of Medicine, Harvard University	2023.08
- BRIC academic webinar	2023.03
- 45 th meeting, The Korean Society of Abdominal Radiology, 2022	2022.06
Diffusion models: foundations and applications in biomedical imaging	
- IEEE International Symposium on Biomedical Imaging (ISBI) 2023	2023.05
Diffusion models for inverse problems	
- Joint Workshop on Machine Learning for Urban Environmental Modeling and Flows	2026.09
- The Korean Institute of Broadcast and Media Engineers	2025.12
- University of Minnesota	2025.11
- LANL	2024.11
- IPA seminar, Korea University	2024.09
- Krafton AI	2024.09
- DRGem	2024.08
- LG AI Research	2024.08
- Twelve Labs	2024.06
- AI SEOUL 2024	2024.02
- Inference & control group seminar, Donders Institute, Radboud Univ.	2023.01
- LANL T-CNLS seminar, 2022	2022.08

Preprints

- [P7] Advancing Ultra Low-Field MRI with Synthetic Data and Deep Learning-Based Image Enhancement for Brain Volume Analysis
Peter Hsu, Elisa Marchetto, [Hyungjin Chung](#), Dohun Lee, Jong Chul Ye, Daniel Sodickson, Jelle Veraart, Patricia Johnson
- [P6] ContextMRI: Enhancing Compressed Sensing MRI through Metadata Conditioning
[Hyungjin Chung*](#), Dohun Lee*, Zihui Wu, Byung-Hoon Kim, Katie Bouman, Jong Chul Ye
- [P4] ACDC: Autoregressive coherent multimodal generation using diffusion correction
[Hyungjin Chung*](#), Dohun Lee*, Jong Chul Ye
- [P3] A survey on diffusion models for inverse problems
Giannis Daras, [Hyungjin Chung](#), Chieh-Hsin Lai, Yuki Mitsufuji, Jong Chul Ye, Peyman Milanfar, Alexandros G Dimakis, Mauricio Delbracio
- [P2] Amortized Posterior Sampling with Diffusion Prior Distillation

Abbas Mammadov*, Hyungjin Chung*, Jong Chul Ye

[P1] Generative AI for Medical Imaging: extending the MONAI Framework

Pinaya *et al.* (Hyungjin Chung: Contributing author)

Patent

US patent application

- Score-based Diffusion Model for Accelerated MRI and Apparatus thereof 2023

Korea patent publication

- Crowd Deep Learning Method of Medical Artificial Intelligence and Apparatus thereof 2025

- Score-based Diffusion Model for Accelerated MRI and Apparatus thereof 2024

- Task-agnostic image processing method and apparatus using transformer and federated split learning 2024

- Tomography image processing method using neural network based on unsupervised learning to remove missing cone artifacts and apparatus therefor 2023

- Two-Stage unsupervised learning method for 3D Time-of-flight MRA reconstruction and the apparatus thereof 2023

Korea patent application

- Accelerating method of conditional diffusion models for inverse problems using stochastic contraction and the apparatus thereof 2021

- Extreme condition reconstruction method HAADF-STEM-EDX tomography using unsupervised deep learning and the apparatus thereof 2021

References

Byung-Hoon Kim

2024.08 — 2025.08

CIO (EverEx)

egyptdj@yonsei.ac.kr

Jong Chul Ye

2019.03 — 2025.02

Ph.D. advisor (KAIST)

jong.ye@kaist.ac.kr

Mauricio Delbracio

2023.07 — 2023.11

Host (Google)

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Peyman Milanfar

2023.07 — 2023.11

Host (Google)

milanfar@google.com

Michael T. McCann

2022.06 — 2022.08

Host (LANL)

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